

Integrated Ovonic Threshold Switching Selector and Resistive Switching Memory 1S1R in Electrodeposited ZnTe Thin Films

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Chalcogenide materials are promising candidates for next generation memories since they can be stacked to integrate both two-terminal non-volatile memory and volatile selector devices for large-scale-integration crossbar arrays. Traditionally, devices based on chalcogenides have been fabricated using vacuum-dependent and high-temperature methods. In this study, the first demonstration of ovonic threshold switching (OTS) and resistive switching (RS) behaviors in zinc telluride (ZnTe) thin films produced via a rapid, cost-effective, and vacuum-free electrodeposition technique is presented. This method also allows for control over ZnTe film composition within the same electrolyte by varying the deposition potentials. These findings reveal that stoichiometric ZnTe thin films exhibit OTS behavior, while Te-rich ZnTe films display RS characteristics. The OTS selectors show robust threshold switching with controllable operating current levels, whereas the RS memory devices demonstrate reliable switching at low voltages, achieving multilevel switching through variations in DC sweeping voltages. A one-selector-one-resistor (1S1R) architecture is successfully implemented by connecting the stoichiometric ZnTe selector in series with the Te-rich ZnTe memory element, thereby validating the potential of OTS as a selector for crossbar applications. This work provides a significant advancement toward constructing stacked structures of memory and selector devices through electrodeposition methods, paving the way for high-density crossbar array applications.

Neumann computing architecture and the limitation of Moore's law for scaling down architectures.^[1] These emerging technologies include ferroelectric random-access memories (FRAMs), magnetic random-access memories (MRAMs), phase-change random-access memories (PCRAMs), and resistive random-access memories (RRAMs).^[2] Among them, RRAMs (also called resistive switching memories or memristors) have been increasingly considered as promising candidates for next generation memory technologies since they combine the high density of dynamic random-access memories (DRAMs) and the high speed of static random-access memories (SRAMs) with the non-volatility of Flash memories.^[3] RRAMs offer not only excellent memory properties such as low power operation, fast switching speeds, and long data retention and endurance capabilities^[4–12] but also enable integration of high density crossbar hybrid RRAM/complementary metal-oxide-semiconductor (CMOS) arrays with highly scalable sizes using their simple two-terminal

metal-insulator-metal (MIM) architecture.^[13,14] Additionally, as they can mimic the analogue switching and learning characteristics of biological neural synapses, RS devices have been interestingly considered as promising options to be implemented

1. Introduction

Over the past decades, several new memory technologies have emerged to overcome the bottleneck of traditional von

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as artificial synapses for neuromorphic computing applications such as artificial vision systems,^[15,16] visual perception nervous systems,^[17,18] reservoir computing systems^[19–22] and self-powered unidirectional neuromorphic perception networks.^[23]

Fabrication RRAM in crossbar arrays enables the effective miniaturization of the memory cells into $4F^2$ (F is the minimum cell size) and can be used to integrate high-density 3D stacked architectures.^[24] However, integration of RRAM cells in high-density 3D crossbar architectures suffer from the key dilemma of sneak-path leakage issue, which arises from unintended leakage currents from neighboring unselected cells.^[24] This issue affects the reading and writing operations for the stored information and becomes increasingly problematic as the crossbar stack size increases.^[25] To reduce the effect of sneak-paths, select devices with high nonlinearity features can be inserted at every junction in series with the memory cells to isolate a selected cell from the sneak-path leakage in the array. Transistors have been inserted as selectors in a 1-transistor-1-resistor (1T1R) design for most emerging memory technologies.^[25,26] However, the three-terminal structure and large footprint of transistors limit their use in high-density 3D scalable crossbar architectures.^[27] Instead, OTS devices, which are two-terminal selectors, offer a more compatible solution. OTS operates based on a current-controlled negative differential resistance mechanism which is primarily associated with specific chalcogenide materials, allowing for efficient switching between low and high conductivity states. This characteristic makes OTS selectors particularly suitable for addressing sneak-path issues without compromising the scalability and 3D stacking capabilities of RRAM cells.^[28]

Numerous examples have been reported on the development of OTS selectors based on chalcogenide materials,^[29] such as SiTe,^[30,31] ZnTe,^[32–34] CuGeS,^[35] As-Te-Ge-Si-N,^[28] Ga-N-Sb-Te,^[36] and Ge₂Sb₂Te₅.^[37] The reported selectors have demonstrated considerable characteristics. These include high selector nonlinearity with low operating currents (<10 fA),^[30,35,37] bidirectional robust threshold switching behavior at temperatures up to 400 °C,^[34,35] 30 nm device size with fast switching speed of less than 2 ns,^[28,38] ultrahigh selectivity of 10^{10} and high endurance performance up to 10^8 cycles.^[36,39] However, despite their favorable characteristics, these selector devices face significant challenges related to deposition methods. Conventionally, selectors have been fabricated using techniques such as radio frequency magnetron sputtering,^[28,30,31,34–37,40] and atomic layer deposition.^[32,39,41,42] These “top-down” approaches typically require vacuum equipment, resulting in high deposition costs and slow processing rates. Furthermore, the elevated thermal budgets associated with these methods can be prohibitive, particularly for applications in flexible electronics, where maintaining low temperatures during fabrication is essential. As a result, there is a pressing need for alternative deposition techniques that can overcome these limitations while ensuring compatibility with advanced electronic applications. Electrodeposition is a well-established deposition technique in the electronic industry that offers a vacuum free, low-temperature deposition and low-cost and fast alternative solution for chalcogenide materials deposition.^[43,44] Several chalcogenide films have been successfully electrodeposited for phase change and resistive switching memory devices, demonstrating com-

petitive switching performance.^[45–48] However, the use of electrodeposition for chalcogenide-based selectors has not yet been reported.

In this work, we report the fabrication of OTS selector and RS memory devices based on electrodeposited ZnTe thin films on Si/SiO₂/TiN substrates. The OTS and RS behaviors are achieved by controlling the Te and Zn ratio within the ZnTe thin films through the application of different deposition potentials during the electrodeposition process. Specifically, stoichiometric ZnTe exhibits OTS behavior, while Te-rich ZnTe demonstrates RS behavior. Both OTS and RS devices showcase competitive switching properties, including good ON/OFF ratio, controllable current level, and multilevel switching behavior. A series connection was established between a stoichiometric ZnTe selector (1S) and a Te-rich ZnTe memory cell (1R) to validate the potential feasibility of employing OTS as a selector in a 1S1R architecture.

2. Results and Discussion

2.1. ZnTe Thin Film Electrodeposition and Characterization

Non-aqueous electrodeposition of ZnTe thin films was carried out in a CH₂Cl₂ and MeCN mixed solution using commercial Zn precursor, Zn(TFSI)₂ and homemade Te precursor of [(n-C₄H₉)₄N]₂[TeCl₆]. Pre-patterned TiN thin films on Si/SiO₂ substrates were used as the deposition electrode. More experimental information on the electrodeposition is provided in the method section. Cyclic voltammograms (CVs) of the supporting electrolyte (0.1 M [Bmpyrd][TFSI]) only was recorded and identified the onset voltage of the electrolyte breakdown to be above −2.0 V versus Ag|AgCl (details in Figure S1, Supporting Information). After confirmation of the breakdown voltage, the individual electrodeposition processes for Zn(TFSI)₂ and [(n-C₄H₉)₄N]₂[TeCl₆] on the TiN surface were investigated before a solution consisting of both precursors was prepared to investigate electrodeposition of ZnTe alloy. CVs for each solution were recorded at 50 mV s^{−1} to gain insights into the phase formation process for Zn and Te, as shown in Figures S2 and S3 (Supporting Information). Detailed analyses are also presented in the Supporting Information. The electrodeposition of Te from [TeCl₆]^{2−} is a complex process involving reduction of Te(IV) and Te(II) chloride complexes and has been discussed in an earlier publication.^[49] To ensure the deposition of ZnTe, a solution containing 5 mM Zn(TFSI)₂ and 300 μM [(n-C₄H₉)₄N]₂[TeCl₆] in 0.1 M [Bmpyrd][TFSI] was prepared. Figure 1a shows the CVs of the three voltammetric cycles at a scan rate of 0.05 V s^{−1} (individual CVs shown in Figure S4, Supporting Information).

The first CV cycle shows three reduction peaks at −0.435 V (E_{1c}), −0.785 V (E_{2c}), and −1.481 V (E_{3c}) versus Ag|AgCl (V), which could be associated with the Te deposition, codeposition of ZnTe compound, and deposition of Zn-rich binary telluride. On the reverse scan, an oxidation (stripping) peak is observed at ca. +0.7 V (E_{1a}) versus Ag|AgCl, which we assign to Zn stripping. The second and the third CV repeats are quite identical but show slight changes to the first CV cycle due to the change of the surface, which was modified during the first cycle. In the second and the third cycle, we noticed a decrease in the current and charge associated with the oxidation peak. Also, the reduction peaks E_{1c}, E_{2c}, and E_{3c} shift to more positive potential where the

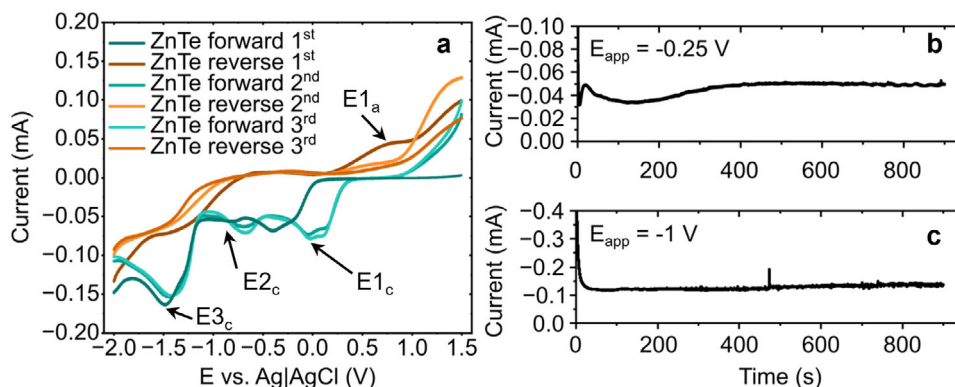


Figure 1. a) Overlaid forward and reverse CV cycles of the Si/SiO₂/TiN substrate in 0.1 M [Bmpyr][TFS] in dry CH₂Cl₂/MeCN (2:1 by vol) containing 5 mM Zn(TFSI)₂ + 300 μM [(n-C₄H₉)₄N]₂[TeCl₆] recorded at a scan rate of 0.05 V s⁻¹. Current–time transient for ZnTe electrodeposition at a deposition potential of b) –0.25 V versus Ag|AgCl, and c) –1 V versus Ag|AgCl.

reduction peak E1_c can be resolved into two peaks. We suggest that Te starts depositing on the surface at potential ca. +0.25 V versus Ag|AgCl and Zn underpotential deposition commences at more negative potential. In the potential region above ca. –1.3 V versus Ag|AgCl, most likely Zn plating is the dominant process and in competition with the formation of Zn-rich binary telluride. Whereas between ca. +0.25 V versus Ag|AgCl and –1 V versus Ag|AgCl both Te, and ZnTe can take place. We then examined two plating potentials, –0.25 V and –1.0 V versus Ag|AgCl to obtain ZnTe thin films with different Zn:Te compositions, i.e., Zn_xTe_y. In both cases, the electrode was held at open-circuit potential for the initial 5 s, followed by applying the predefined potentials at the working electrode for 900 s, as shown in the current transients in Figure 1b,c.

The composition of the electrodeposited ZnTe thin films was characterized by scanning electron microscopy (SEM) equipped with energy-dispersive X-ray spectroscopy (EDX). Top view SEM image of the ZnTe thin film electrodeposited at –0.25 V is shown in Figure 2a. The image shows that the film is deposited across the entire substrate. EDX mappings in Figure 2b,c suggest that

both Zn and Te elements are deposited. The EDX spectrum of this film is plotted in Figure S5a (Supporting Information). The Zn:Te composition ratio of this film is established to be 51:49, which is close to the stoichiometric value. On the other hand, Figure 2d–f presents the SEM and EDX mapping of the ZnTe film deposited at –1.0 V. The EDX spectrum of this film is plotted in Figure S5b (Supporting Information). A significant variation of the film morphology can be observed. Such change of morphology is also accompanied by the change of composition to a Zn:Te composition ratio of 17:83, indicating a Te-rich film. The change over film composition via deposition voltage highlights the control electrodeposition can offer in the growth of chalcogenide films and proves to be crucial to realize the different switching functionalities presented later in this paper. X-ray photoelectron spectroscopy (XPS) measurements were also performed on these two films, further confirming their compositions (shown in Figure S6, Supporting Information). Details of the XPS analysis can also be found in the Supporting Information. We will therefore refer to these two films as stoichiometric ZnTe and Te-rich ZnTe in the following parts of this work.

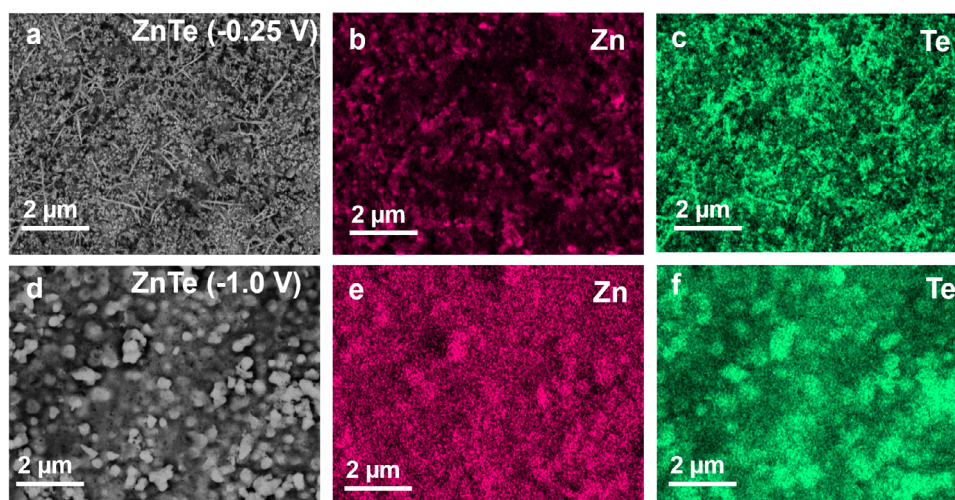


Figure 2. SEM image showing the ZnTe film electrodeposited at a) –0.25 V, and d) –1.0 V. b,c,e,f) EDX element maps showing the deposition of Zn and Te across the substrate.

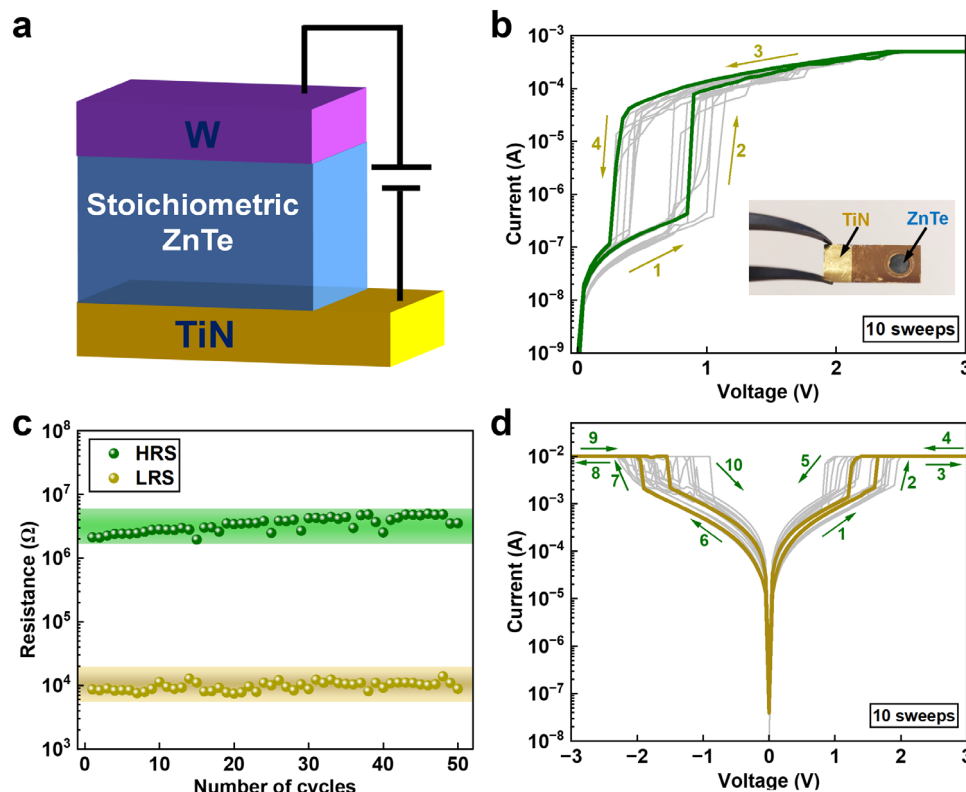


Figure 3. Electrical characteristics of stoichiometric ZnTe-based OTS selectors. a) Schematic of stoichiometric ZnTe thin film-based OTS selector probed between TiN and W electrodes, TiN/ZnTe/W. b) Consecutive 10 DC-IV sweeps of the OTS selector, representing a typical volatile RS behavior. The inset is a photo image of the sample. c) DC endurance characteristics. d) Consecutive 10 DC-IV sweeps of the OTS selector after forming at high CC level.

Further work is needed to improve the uniformity and purity of the ZnTe films.

2.2. Electrical Characteristics of Electrodeposited ZnTe OTS Selector

In this section, we demonstrate the development of OTS selector based on the electrodeposited stoichiometric ZnTe thin film using a titanium nitride (TiN) layer as the bottom contact and tungsten (W) as the top electrode (schematic shown in Figure 3a). The inset in Figure 3b shows a photograph of the sample after fabrication. The as-electrodeposited ZnTe selector was initially in the high resistance state (HRS). An electroforming process using a compliance current (CC) of 100 μ A was required to initialize the selector, as shown in Figure S7a (Supporting Information). After initialization, the selector demonstrates a typical threshold switching behavior upon sweeping the selector with a DC bias between 0 and 3 V and a CC of 500 μ A. The device current increased gradually with voltage until reaching a threshold voltage (V_{th}) of about 0.9 V. This was followed immediately by an abrupt current transition with more than two orders of magnitude, switching the device to the low resistance state (LRS), as shown in Figure 3b. However, upon reverse sweeping from 3 V to 0 V, the device returned to its HRS when the voltage was below the holding voltage (V_h) of about 0.5 V. Subsequent 9 DC-IV cycles (gray curves) are similar to the initial switching cycle (green curve). This indicates

the volatile threshold behavior since the LRS can only be maintained if the applied voltage is higher than the V_h . Figure 3c exhibits a reproducible cycling endurance in the DC sweep mode for 50 cycles. Both the HRS and LRS showed good uniformity without significant degradation, with a ratio between the two resistance states of more than two orders of magnitude. To elucidate further the switching stability of the selector, cumulative probability distributions for the HRS and the LRS are shown in Figure S8a (Supporting Information). The plot indicates that both states are uniform. Moreover, good uniformities can also be seen for both V_{th} and V_h (shown in Figure S8b, Supporting Information).

Moreover, the operation current level of the OTS selector can be controlled via the electroforming CC level. Figure S7b (Supporting Information) presents the forming of the device at high CC level of 1 mA. Figure 3d shows the subsequent OTS behavior of the selector with the direction of the current sweep indicated by arrows. Forming the selector at a high CC level decreases the resistance of the HRS, which resulted in a lower ON/OFF ratio between the HRS and LRS (about 1 order of magnitude) in comparison with the forming of the device at a lower CC. Similar OTS behavior was also observed from a device consisting of the stoichiometric ZnTe film sandwiched between TiN top and bottom contacts (Figure S9, Supporting Information), suggesting that this OTS behavior is independent of the electrode. We further investigated the current conduction mechanism of the OTS device where the space-charge-limited-current (SCLC) model presents

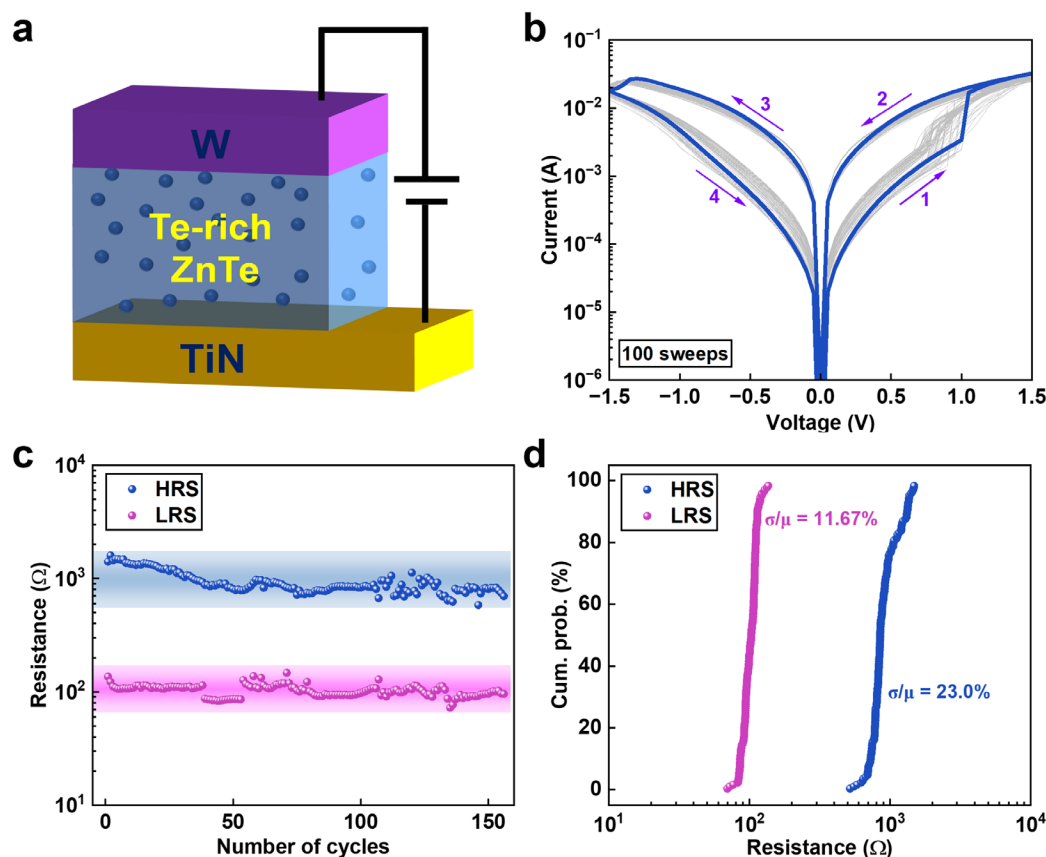


Figure 4. Electrical characteristics of Te-rich ZnTe-based RS memory. a) Schematic of Te-rich ZnTe thin film-based RS memories probed between TiN and W electrodes, TiN/ZnTe/W. b) Consecutive I - V characteristics of the RS memory, representing a typical bipolar switching. c) DC endurance characteristics. The HRS and LRS were read @ 0.5 V. d) Cumulative probability of the HRS and LRS.

the best agreement to our experimental results (shown in Figure S10, Supporting Information). More discussions of the switching and conduction mechanism are in the Supporting Information. Based on the experimental results, a possible operation mechanism for such CC controllable OTS behaviors is discussed in the Supporting Information and schematically shown in Figure S11 (Supporting Information).

2.3. Electrical Characteristics of Electrodeposited ZnTe RS Memory

We now focus on the electrical switching properties of the Te-rich ZnTe thin film-based device (schematic shown in Figure 4a). The device was initially in a HRS and a forming process was required to initialize the devices for the resistive switching effect, as shown in Figure S13a (Supporting Information). Note that a CC of 10 mA was used during the forming process to limit the operating current and to protect the device from breakdown. After forming the device can switch between HRS and LRS. One hundred consecutive DC-IV curves showing typical bipolar and reproducible switching behavior of the device are presented in Figure 4b with the direction of the current sweep indicated by arrows. Upon sweeping between ± 1.5 V, the device switched from HRS to LRS (SET process) using a positive DC sweep from 0 to

1.5 V. Unlike the OTS selector presented above, the Te-rich ZnTe based device requires a negative DC sweep from 0 to -1.5 V to switch it back to the HRS (RESET process), showing a typical bipolar resistive switching behavior. It is worth highlighting that the switching is compliance free, which is desirable for simple-to-make architecture and energy efficient peripheral circuitry.^[50] In addition, the device can also be SET at negative polarity and RESET at positive polarity (as shown in Figure S13c, Supporting Information), manifesting a non-polar switching characteristic. Figure 4c shows a reproducible cycling endurance in the DC sweep mode for over 150 cycles. The resistance window between the HRS and LRS is about one order of magnitude at 0.1 V, which is sufficient for memory applications. Both the resistance states exhibited good uniformity without significant decay. To further elucidate the switching stability of the device, cumulative probability distributions for the HRS and LRS are plotted in Figure 4d. The figure clearly shows that the HRS and LRS are uniformly distributed with the coefficients of variation (σ/μ , σ is the standard deviation and μ is the mean value) calculated to be 23.0% and 11.7% for the HRS and LRS, respectively. Additionally, high uniformities were also seen for both V_{SET} and V_{RESET} as illustrated in Figure S13d (Supporting Information), where the coefficients of the variation were 4.3% and 0.8%, respectively. The high uniformities of both programming voltages are essential for large-scale RRAM applications.

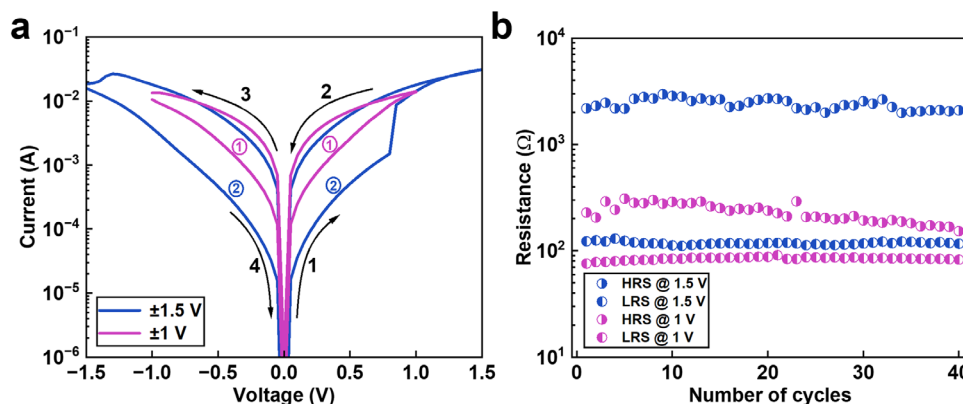


Figure 5. Multilevel RS behavior of the Te-rich ZnTe thin film-based RS memory. a) DC-IV characteristics showing the multilevel RS behavior induced by varying the applied voltage sweep. b) DC endurance of the device showing the reproducible multilevel RS behavior under different DC sweep voltages. The HRS and LRS were read @ 0.1 V.

The RS properties in our Te-rich ZnTe thin film-based RS memory can be controlled by varying the applied voltage sweep to achieve a multilevel RS behavior. **Figure 5a** exhibits the I - V characteristics of a RS memory using different DC sweep voltages (± 1 V and ± 1.5 V). After the forming process, the device can be switched between the HRS and LRS upon the application of a negative and positive DC sweep, respectively. During the first RESET DC sweep from 0 to -1 V (pink curve), the RESET was observed at ca. -1 V and the HRS was measured to be ca. 300Ω (@ 0.1 V). **Figure S13b** (Supporting Information) shows consecutive 40 DC-IV curves of the device. After the device was SET back to the LRS, the subsequent RESET was conducted with a DC sweep from 0 to -1.5 V (blue curve). The RESET during this sweep occurred at ca. -1.5 V and resulted in an HRS of ca. 3000Ω (@ 0.1 V), which is 10 times higher than the previous HRS. These results clearly show that the device switches to different HRS level, while the LRS remained almost the same. Therefore, the device demonstrates three distinct resistance state levels including one LRS and two HRS. The multilevel RS behavior is uniform and reproducible for over 40 cycles, as presented in **Figure 5b**. The multilevel RS behavior is likely to be induced by a combination of elongated electrical stress and potentially larger electrical field applied on the device, as previously observed in our crossbar GeSbTe-based memory devices.^[48] Such behavior suggests our RS memory can be considered as a promising candidate electronic synapse for neuromorphic system and multilevel data storage applications.

To investigate the conduction mechanism in the TiN/Te-rich ZnTe/W RS memory, the positive and negative parts of the I - V curve were replotted in double-logarithmic scale as shown in **Figure S14a,b** (Supporting Information), respectively. The fitting result shows that the Ohmic conduction mechanism dominates in the LRS state, while the HRS state is predominately a combination of Ohmic conduction mechanism at low applied voltage region, and SCLC at a high applied voltage region. A similar conduction mechanism was previously reported in chalcogenide GeSbTe-based RRAM devices.^[46–48] In addition, a possible RS mechanism based on the electrical results of the electrodeposited Te-rich ZnTe thin film-based RS memory is proposed in **Figure**

S15 (Supporting Information) and discussed in the Supporting Information.

2.4. Electrical Characteristics of Electrodeposited ZnTe Based 1S1R Structure

The 1S1R configuration is typically used in crossbar architectures to reduce the sneak-paths effect. Here, we demonstrate the 1S1R structure by electrically connecting the electrodeposited stoichiometric ZnTe OTS selector (1S) and Te-rich ZnTe RS memory (1R) in series, as shown **Figure 6a**. **Figure 6b** exhibits the representative DC-IV curves for each selector (yellow curve) and RS memory (blue curve). The selector switches at a V_{th} of 1.5 V and V_h of 1 V at positive polarity and -1.5 and -1 V at negative polarity, respectively. Reproducible DC-IV switching characteristics for the selector are shown in **Figure S16a** (Supporting Information). The RS memory shows a bipolar RS with a SET/RESET process occurring at negative/positive voltage polarity. Reproducible DC-IV switching characteristics for the RS memory are shown in **Figure S16b** (Supporting Information). The I - V characteristic of the 1S1R structure is shown in **Figure 6c** with the steps indicated by arrows and numbers. Initially, both the memory and selector devices were at HRS (logic “0”) state. Upon the negative DC sweep, the potential drop was predominately on the selector initially due to its higher resistance compared with the memory (step 1). As the applied voltage increased, the selector firstly turned to LRS when the threshold voltage V_{th} (ca. -1.6 V) was reached. This was quickly followed by the SET process of the memory into LRS (logic “1”) as now most of the potential drop was on the memory (step 2). When the voltage was swept back to zero, both memory and selector remained in the LRS until the selector turned back to HRS at below hold voltage V_h of ca. -0.9 V (step 3). However, the memory remained at LRS (logic “1”) due to its non-volatile behavior (step 4).

In the subsequent RESET process, the 1S1R device was biased from 0 to 3 V. Within the low positive bias region, the I - V curve of the 1S1R followed the behavior of the selector because the memory was in LRS (step 5). As the applied bias continued to increase, the selector switched ON into LRS at ca. 1.0 V (step 6).

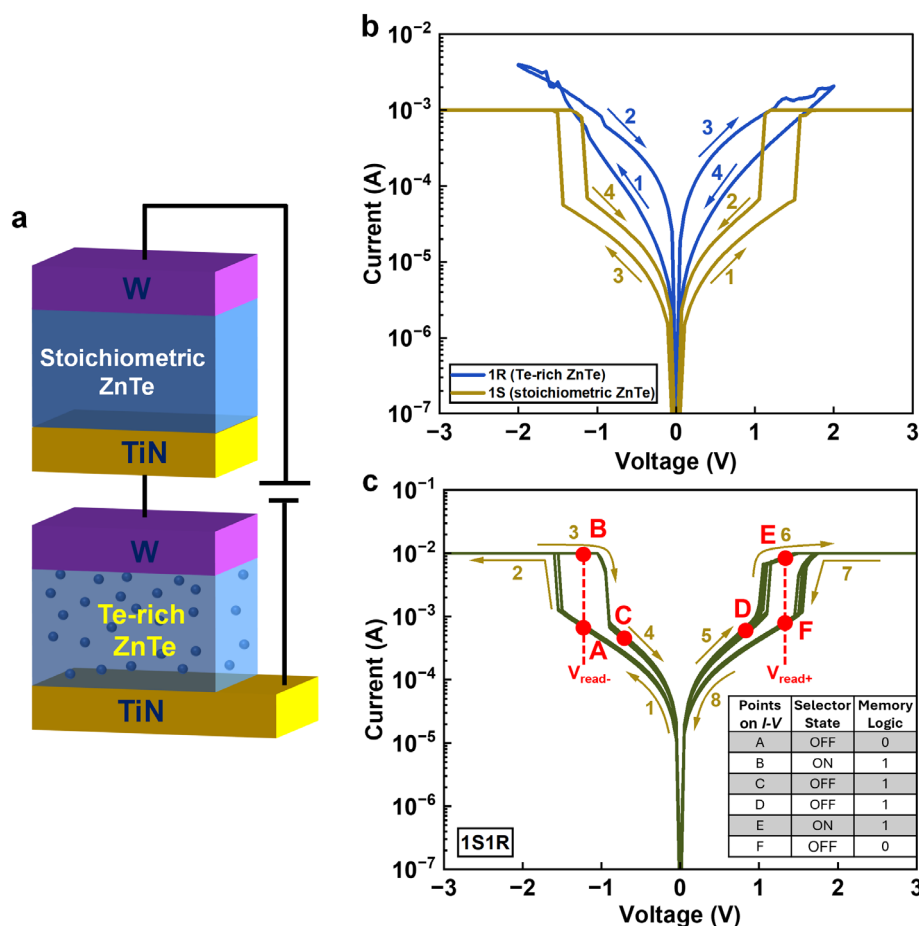


Figure 6. Electrical characteristics of 1S1R structure based on electrodeposited ZnTe films. a) Schematic of a stoichiometric ZnTe film-based selector and Te-rich ZnTe film based RS memory connected in series through the electrical wire connection. b) I - V switching curves of the selector (yellow curve) and memory (blue curve). c) Successive 20 DC-IV curves of the 1S1R structure with arrows indicating the switching steps. The inset table lists the logic states of selector and memory at different points on the I - V curve.

Subsequently, the memory underwent a RESET process to HRS (logic “0”) at ca. 1.8 V (step 7). The operation cycle was then completed by switching the selector back to HRS when the bias was swept from 3 back to 0 V. The states for both selector and memory at different points (A–F) in the I - V curve are tabulated in the inset table in Figure 6c, and the 1S1R device can be read at both polarities ($V_{\text{read-}}$ and $V_{\text{read+}}$). 20 successive DC-IV sweeps of the 1S1R device are presented in the figure with a further 50 sweeps shown in Figure S16c (Supporting Information). These results clearly demonstrate the proper operation of 1S1R cell by integrating the stoichiometric ZnTe selector and Te-rich ZnTe RS memories from our electrodeposition process and provide a scientific foundation for further understanding and enhancing device performance in future work. Further works will also focus on evaluating their performance in crossbar structures which can also be realized via the electrodeposition approach.^[45,48]

3. Conclusions

In summary, we have successfully demonstrated a typical OTS selector and bipolar RS memory using ZnTe thin films fabricated

through electrodeposition. The OTS selector device based on stoichiometric ZnTe film exhibits threshold switching behavior in which the operating current can be controlled by the CC level upon the forming process. The OTS performance as a feasible selector is promising in terms of its ratio, switching uniformity and endurance. On the other hand, the Te-rich ZnTe thin film exhibits bipolar non-volatile RS memory behavior. The RS memory device shows a reproducible switching effect. Furthermore, by integrating both the OTS and memory devices, we successfully established a functional 1S1R with commendable switching performance. These experimental findings, combined with our ability to control the composition of ZnTe films simply by varying the deposition potential within a single electrolyte, underscore the exciting potential of using electrodeposition techniques for fabricating advanced 1S1R architectures.

4. Experimental Section

Electrolytes, Materials, and Chemicals: Electrolytes were prepared in a mixed solvent system consisting of anhydrous acetonitrile (Sigma-Aldrich, 99.8%, MeCN) and dichloromethane, CH_2Cl_2 (Sigma-Aldrich, 95%), dried

and degassed by refluxing with CaH_2 , followed by distillation and then stored in a glovebox.

ZnTe electrodeposition solutions were prepared from a mixture of Zn and Te salts; dry zinc di[bis(trifluoromethylsulfonyl)imide] ($\text{Zn}(\text{TFSI})_2$, 99.5%, Solvionic), dry tetrabutylammonium hexachlorotelluride $[(n\text{-C}_4\text{H}_9)_4\text{N}]_2[\text{TeCl}_6]$, and with dry 1-butyl-1-methylpyrrolidinium bis(trifluoromethanesulfonyl)imide (99.9%, Solvionic, [Bmpyrd][TFSI]) as a supporting electrolyte dissolved in a mixed solvent made of 10 mL CH_2Cl_2 and 5 mL MeCN at room temperature.

Electrochemical experiments were carried out using a solution of mixed-salts system, prepared using various ratios/concentrations of $\text{Zn}(\text{TFSI})_2$ and $[(n\text{-C}_4\text{H}_9)_4\text{N}]_2[\text{TeCl}_6]$, but keeping the concentration of the supporting salt [Bmpyrd][TFSI] fixed at 0.1 M. All solutions were prepared in N_2 filled glovebox using dry glassware.

Electrodeposition of ZnTe Films: The experiments were performed using the following electrodes: i) silicon substrates with pre-deposited TiN bottom electrodes (4 mm in diameter) serve as the working electrode for electrodeposition. TiN bottom contacts (200 nm thick) were sputtered on Si/SiO_2 substrates by medium frequency plasma assisted magnetron sputtering (Leybold Optics HELIOS Pro XL) at room temperature.^[51,52] This process combines reactive middle frequency sputtering with an additional RF plasma source. During each rotation of the plate holding the substrate, a thin layer of Ti was deposited from dual magnetron metal targets (99.99% purity) using a power of 3000 W in an Ar atmosphere. The thin film was then transformed into a nitride layer by passing the substrate underneath the N_2 plasma of the RF source. The N_2 and Ar flow rates were maintained at 30 and 35 sccm, respectively and a high drive speed of 180 rpm was applied to enhance the film uniformity. The deposition rate was 0.161 nm s^{-1} . ii) leak-free (LF-2-45, Alvatek), Ag/AgCl wire (0.1 M $[\text{N}^n\text{Bu}_4]\text{Cl}$ in anhydrous CH_2Cl_2), serve as the reference electrode and iii) a disc of Pt/Ir (90%:10%) alloy serve as the counter electrode. In general, all electrodes were contained in a one compartment homemade cell where the distance between the working electrode and the counter electrode is always the same and without separating the reference or the counter electrodes.

All cyclic voltammetry (CV) and electrodeposition experiments were carried out in a recirculating glovebox (Belle) to exclude moisture and air. Oxygen levels were kept below 10 ppm. A microAutolab 3 potentiostat and a Nova 1 software package were used for all electrochemical measurements. Prior to ZnTe electrodeposition, voltammetry was recorded on a TiN bottom contact in 0.1 M [Bmpyrd][TFSI] electrolyte containing 5 mM $\text{Zn}(\text{TFSI})_2$ and 300 μM $[(n\text{-C}_4\text{H}_9)_4\text{N}]_2[\text{TeCl}_6]$ recorded at a scan rate of 0.05 V s^{-1} .

Morphology, Composition, and Electrical Characterization: Scanning electron microscopy (SEM) was performed using Zeiss Sigma 500 VP FE-SEM. Energy dispersive X-ray spectroscopy (EDX) was performed using an Oxford Instruments Ultim 170 EDS system. X-ray diffraction (XRD) was performed using a Rigaku SmartLab diffractometer ($\text{Cu-K}\alpha$) with a grazing incidence angle of 1° , and Hypix 3000 detector in 1D mode. The electrical characteristics were measured at room temperature and ambient pressure using a tungsten (W) probe as a top electrode connected to a Keysight (B1500) system. A 200 nm thick of TiN sputtered on the ZnTe film through a shadow mask containing 250 μm diameter circles as a top contact was also used. For all measurements, the voltage was applied to the top contact while the bottom electrode was grounded.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are openly available in University of Southampton repository at <https://doi.org/10.5258/SOTON/D3405>, reference number 3405.

Keywords

1S1R, electrodeposition, ovonic threshold switching, resistive switching, ZnTe

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